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Practical week - 6

routes/catalog.py :

```
from fastapi import APIRouter, HTTPException

router = APIRouter()

books = [

    {

        "id": 101,

        "title": "Python Programming",

        "author": "John Smith",

        "price": 450,

        "isbn": "9781234567890",

        "status": "Available"

    },

    {

        "id": 102,

        "title": "Database Management",

        "author": "Robert Brown",

        "price": 550,

        "isbn": "9789876543210",

        "status": "Available"

    }

]

@router.get("/books") def

get_books():

    return books


@router.get("/books/{book_id}")

def get_book(book_id: int):

    for book in books:

        if book["id"] == book_id:

            return book raise
```

```
HTTPException(  
    status_code=404,  
    detail="Book not found"  
)
```

routes/admin.py :

```
from fastapi import APIRouter, HTTPException  
from pydantic import BaseModel, Field  
from routes.catalog import books  
router = APIRouter()
```

```
class Book(BaseModel):
```

```
    title: str  
    author: str  
    price: float = Field(ge=0)  
    isbn: str  
    status: str
```

```
@router.post("/admin/books") def
```

```
add_book(book: Book):  
    new_id = max([b["id"] for b in books], default=100) + 1  
    new_book = {  
        "id": new_id,  
        book.model_dump()  
    }  
    books.append(new_book)  
    return {  
        "message": "Book added successfully",  
        "book": new_book  
    }
```

```
@router.put("/admin/books/{book_id}")
```

```
def update_book(book_id: int, book: Book):  
    for index, existing_book in enumerate(books):  
        if existing_book["id"] == book_id:
```

```

        updated_book = {
            "id": book_id,
            book.model_dump()
        }
        books[index] = updated_book
    return {
        "message": "Book updated successfully",
        "book": updated_book
    }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

@router.delete("/admin/books/{book_id}")
def delete_book(book_id: int):
    for index, book in enumerate(books):
        if book["id"] == book_id:
            deleted_book = books.pop(index)
            return {
                "message": "Book deleted successfully",
                "book": deleted_book
            }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

```

main.py :

```

from fastapi import FastAPI
from routes.catalog import router as catalog_router
from routes.admin import router as admin_router
app = FastAPI(
    title="BookFlow Library API"
)
app.include_router(catalog_router)
app.include_router(admin_router)

```

GET default all the data:

default

GET /books Get Books

Parameters

No parameters

Execute

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8080/books' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8080/books

Server response

Code Details

200

Response body

{\n \"id\": 101,\n \"title\": \"Python Programming\",\n \"author\": \"John Smith\",\n \"price\": 499,\n \"isbn\": \"9781234567890\",\n \"status\": \"Available\"\n},\n{\n \"id\": 102,\n \"title\": \"Database Management\",\n \"author\": \"Robert Brown\",\n \"price\": 399,\n \"isbn\": \"9789876543210\",\n \"status\": \"Available\"\n}\n}

Response headers

content-length: 240\ncontent-type: application/json\ndate: Tue, 15 Sep 2026 06:05:14 GMT\nserver: uniform

GET:

GET /books/{book_id} Get Book

Parameters

Cancel

Name Description

book_id * required integer (path) 101

Execute

Response:

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8080/books/101' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8080/books/101

Server response

Code Details

200

Response body

{\n \"id\": 101,\n \"title\": \"Python Programming\",\n \"author\": \"John Smith\",\n \"price\": 499,\n \"isbn\": \"9781234567890\",\n \"status\": \"Available\"\n}

Response headers

content-length: 117\ncontent-type: application/json\ndate: Tue, 15 Sep 2026 06:22:08 GMT\nserver: uniform

POST:

POST

/admin/books Add Book

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Java Programing",
  "author": "Sujit",
  "price": 600,
  "isbn": "9781234567891",
  "status": "Available"
}
```

Execute

Response:

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/admin/books' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Java Programing",
    "author": "Sujit",
    "price": 600,
    "isbn": "9781234567891",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8000/admin/books

Server response

Code	Details
200	Response body <pre>{ "message": "Book added successfully", "book": { "id": 10, "title": "Java Programing", "author": "Sujit", "price": 600, "isbn": "9781234567891", "status": "Available" } }</pre> Response headers <pre>content-length: 160 content-type: application/json date: Tue, 15 Sep 2026 06:34:18 GMT server: uvicorn</pre>

PUT:

PUT

/admin/books/{book_id} Update Book

Parameters

Name	Description
book_id * required	101
integer (path)	

Request body required

Edit Value | Schema

```
{
  "title": "JavaScript Essentials",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
}
```

Response:

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8080/admin/books/101' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "JavaScript Essentials",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:8080/admin/books/101

Server response

Code

Details

200

Response body

```
{
  "message": "Book updated successfully",
  "book": {
    "id": 101,
    "title": "JavaScript Essentials",
    "author": "John Smith",
    "price": 450,
    "isbn": "9781234567890",
    "status": "Available"
  }
}
```

Response headers

```
content-length: 169
content-type: application/json
date: Tue, 15 Sep 2026 06:28:50 GMT
server: uvicorn
```

DELETE:

DELETE /admin/books/{book_id} Delete Book

Parameters

Cancel

Name	Description
book_id * required	
integer	
(path)	
102	

Escape

Response:

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/admin/books/102' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8080/admin/books/102

Server response

Code

Details

200

Response body

```
{
  "message": "Book deleted successfully",
  "book": {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9789876543210",
    "status": "Available"
  }
}
```

Response headers

```
content-length: 167
content-type: application/json
date: Tue, 15 Sep 2026 06:30:35 GMT
server: uvicorn
```